

Ryan Cho

Senior Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Full stack engineer with 7+ years building secure, audit-ready platforms in banking and asset management. Deep experience in Python/FastAPI, React/TypeScript, PostgreSQL, Redis workers, and AI-assisted workflows. Delivered large API surfaces, schema migrations, RBAC, and compliance reporting with Docker/AWS operations and strong partnership across risk, legal, and engineering teams.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Owned a **FastAPI backend** spanning 200+ API routes for portfolio analytics, control tracking, and regulatory reporting workflows used daily across operations, risk, and audit functions.
- Built **React and TypeScript** product modules with React Query and Vite, enabling evidence review, finding triage, and exportable audit packages across a broad, state-intensive frontend surface.
- Designed **PostgreSQL models** with SQLAlchemy and Alembic migrations, adding JSONB metadata fields, deduplication constraints, and indexed access paths that preserved integrity across high-volume compliance datasets.
- Implemented **RBAC, authentication, and immutable audit logging** with secret handling, input validation, and session controls aligned to banking security standards and defensible provenance requirements.
- Operated **Redis-backed async workers** for document extraction, scanner ingestion, normalization jobs, and scheduled reconciliations, improving throughput for recurring compliance processing pipelines.
- Extended an **AI provider abstraction** for document review and control-mapping assistance, enforcing scoped context, citation guardrails, and human-review checkpoints rather than direct vendor coupling.
- Maintained **Docker Compose and AWS deployment** stacks with managed PostgreSQL, caches, object storage, health checks, and CI gates for linting, type checking, migration safety, and pytest suites.
- Partnered with risk, legal, and engineering leaders to ship **vulnerability remediation, POA&M, and control-mapping capabilities** that translated framework requirements into adopted platform workflows.
- Mentored contributors on **OpenAPI contract discipline**, SDK drift detection, and full-stack debugging across frontend state, worker queues, schema evolution, and release configuration mismatches.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Built **Python services** for trade surveillance and regulatory reporting pipelines that processed high-volume banking transactions under strict reconciliation and data-integrity controls.
- Delivered **React operational dashboards** surfacing workflow exceptions, attestation status, and audit trails for technology risk teams monitoring distributed banking platforms.
- Maintained **PostgreSQL schemas** with normalized tables, Alembic-style migration routines, and validation checks supporting SOX-aligned document review and reporting cycles.
- Enforced **authentication middleware and RBAC rules** across multi-team tooling, including immutable audit logs used during internal investigations and regulatory examinations.
- Integrated **vendor GRC and risk APIs**, mapping findings, evidence artifacts, and remediation states into centralized tracking workflows for technology control owners.
- Strengthened **CI/CD quality gates** with pytest coverage, static analysis, and deployment runbooks that reduced defects reaching production reporting services.
- Collaborated with audit and technology risk partners to produce **control documentation and export packages** clarifying data lineage, security boundaries, and attestation evidence.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Shipped **React and TypeScript** checklist workflows for a SaaS compliance product, improving form validation and analyst usability across client-facing operations screens.
- Extended **Python REST endpoints** for permissions, template configuration, and JSON-driven checklist logic consumed by web and mobile application clients.
- Authored **pytest suites** for API contracts and persistence helpers, catching contract regressions before releases across an expanding service footprint.
- Tuned **PostgreSQL queries and seed migrations** that stabilized summary reporting views as tenant activity increased during platform growth.
- Participated in **code reviews and secure credential practices**, gaining exposure to component architecture and cloud-hosted deployment conventions.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Built **responsive web pages** for a nonprofit foundation, improving content publishing flows and accessibility for community program visitors across devices.
- Connected **third-party donation and newsletter APIs** with reliable JSON exchange, retry handling, and client-side error messaging for fundraising campaigns.
- Improved **frontend load performance** through asset optimization and caching on program landing pages visited during outreach peaks.
- Translated **stakeholder requirements** into maintainable HTML, CSS, and JavaScript modules supported by concise deployment documentation.
- Supported **production troubleshooting** and content updates during launch, establishing version-control and rollback habits for ongoing site maintenance.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, Vite, React Router, React Query, HTML5, CSS3, Component Architecture, State Management

Backend: Python, FastAPI, Pydantic, SQLAlchemy, Alembic, Async Python, pytest, REST APIs, OpenAPI SDKs

Databases: PostgreSQL, JSON/JSONB, Schema Design, Migrations, Indexing, Data Normalization, Redis

Cloud & DevOps: Docker, Docker Compose, AWS, Terraform, CI/CD, IAM, Object Storage, Container Deployment

Monitoring & Tools: Observability, Health Checks, Audit Logs, Linting, Type Checking, Debugging, Git

Workflow & PM: Agile, Cross-functional Collaboration, Technical Documentation, Stakeholder Management, Mentorship